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SDLC Laboratory

Quality Laboratory Manual

Experiment No. 02

To Understand the requirement engineering tasks in Software Development

Course Instructor –

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Experiment No. 02

Title of Experiment: To Understand the requirement engineering tasks in software development

Aim of Experiment: To study and understand the requirement engineering process and its tasks for gathering, analyzing, documenting, validating and managing software requirements

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- PDF viewer/ Office suite (MS Word/ LibreOffice)
- Web Browser (for reference)

Experiment Objectives:

- To understand the concept of requirement engineering
- To identify different requirement engineering tasks
- To learn how requirements are elicited and analyzed
- To understand the importance of software requirement specification (SRS)
- To recognize challenges in managing requirements

Experiment Outcomes:

After completing the experiment, students will be able to –

- Explain requirement engineering and its importance
- Identify major requirement engineering tasks and their deliverables
- Differentiate between functional and non-functional requirements
- Relate requirement engineering to real-world software projects

Theory:

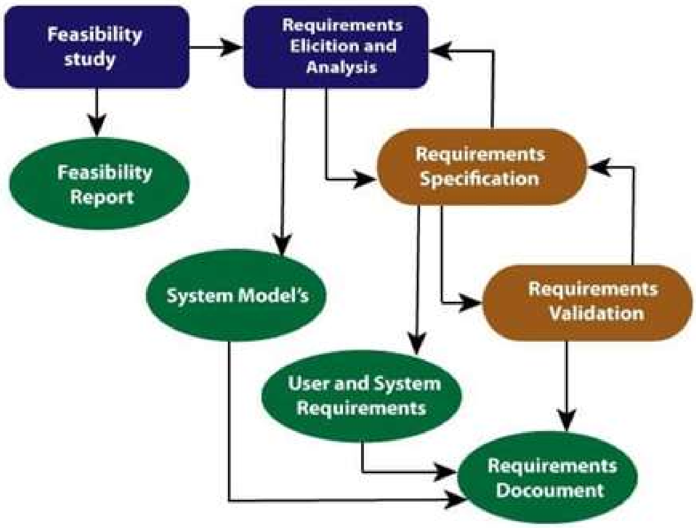
Requirements engineering (RE) refers to the process of defining, documenting, and maintaining requirements in the engineering design process. Requirement engineering provides the appropriate mechanism to understand what the customer desires, analyzing the need, and assessing feasibility, negotiating a reasonable solution, specifying the solution clearly, validating the specifications and managing the requirements as they are transformed into a working system. Thus, requirement engineering is the disciplined application of proven principles, methods, tools, and notation to describe a proposed system's intended behavior and its associated constraints

Requirement Engineering Process

It is a four-step process, which includes –

1. Feasibility Study
2. Requirement Elicitation and Analysis
3. Software Requirement Specification

4. Software Requirement Validation
5. Software Requirement Management



1. **Feasibility Study:**
The objective behind the feasibility study is to create the reasons for developing the software that is acceptable to users, flexible to change and conformable to established standards.

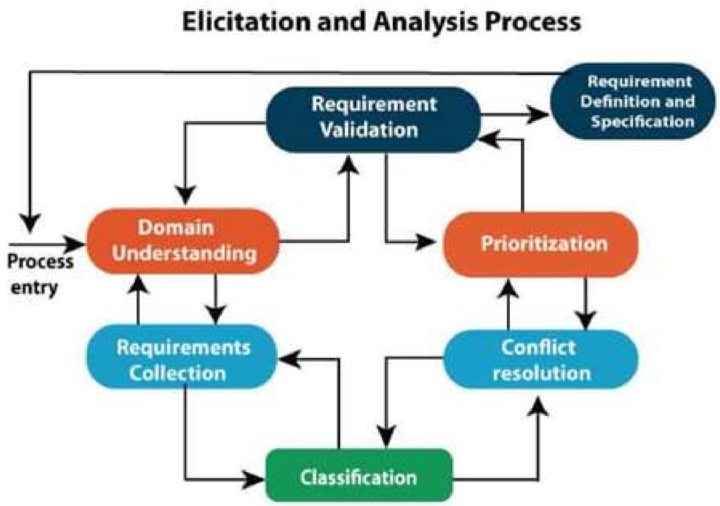
Types of Feasibility:

a. **Technical Feasibility** - Technical feasibility evaluates the current technologies, which are needed to accomplish customer requirements within the time and budget
b. **Operational Feasibility** - Operational feasibility assesses the range in which the required software performs a series of levels to solve business problems and customer requirements
c. **Economic Feasibility** - Economic feasibility decides whether the necessary software can generate financial profits for an organization

2. **Requirement Elicitation and Analysis:**
This is also known as the gathering of requirements. Here, requirements are identified with the help of customers and existing systems processes, if available
Analysis of requirements starts with requirement elicitation. The requirements are analyzed to identify inconsistencies, defects, omission, etc. We describe requirements in terms of relationships and also resolve conflicts if any

Problems of Elicitation and Analysis

- Getting all, and only, the right people involved
- Stakeholders often don't know what they want
- Stakeholders express requirements in their terms
- Stakeholders may have conflicting requirements
- Requirement change during the analysis process
- Organizational and political factors may influence system requirements



3. **Software Requirement Specification:**
Software requirement specification is a kind of document which is created by a software analyst after the requirements collected from the various sources - the requirement received by the customer written in ordinary language. It is the job of the analyst to write the requirement in technical language so that they can be understood and beneficial by the development team
The models used at this stage include ER diagrams, data flow diagrams (DFDs), function decomposition diagrams (FDDs), data dictionaries, etc.
   - **Data Flow Diagrams**: Data Flow Diagrams (DFDs) are used widely for modeling the requirements. DFD shows the flow of data through a system. The system may be a company, an organization, a set of procedures, a computer hardware system, a software system, or any combination of the preceding. The DFD is also known as a data flow graph or bubble chart
   - **Data Dictionaries**: Data Dictionaries are simply repositories to store information about all data items defined in DFDs. At the requirements stage, the data dictionary should at least define customer data items, to ensure that the customer and developers use the same definition and terminologies
   - **Entity-Relationship Diagrams**: Another tool for requirement specification is the entity-relationship diagram, often called an "E-R diagram." It is a detailed logical representation of the data for the organization and uses three main constructs i.e. data entities, relationships, and their associated attributes

4. **Software Requirement Validation:**
After requirement specifications developed, the requirements discussed in this document are validated. The user might demand illegal, impossible solution or experts may misinterpret the needs. Requirements can be the check against the following conditions –
   - If they can practically implement
   - If they are correct and as per the functionality and specially of software
   - If there are any ambiguities
   - If they are full
   - If they can describe

Requirements Validation Techniques
Requirements reviews/inspections: systematic manual analysis of the requirements

- Prototyping: Using an executable model of the system to check requirements
- Test-case generation: Developing tests for requirements to check testability
- Automated consistency analysis: checking for the consistency of structured requirements descriptions

5. **Software Requirement Management:**
Requirement management is the process of managing changing requirements during the requirements engineering process and system development. New requirements emerge during the process as business needs a change, and a better understanding of the system is developed. The priority of requirements from different viewpoints changes during development process. The business and technical environment of the system changes during the development

Task to Perform

Case Study: Student Feedback Management System

An engineering institute plans to develop a Student Feedback Management System to collect feedback from student about courses, faculty and infrastructure, generate analytical reports and support continuous improvement

Task for students

Analyze the above case study and perform the following requirement engineering activities –

- Feasibility study
  - Identify technical and operational feasibility factors
- Requirement elicitation
  - List at least 5 functional requirements
  - List at least 2 non-functional requirements
- Requirement analysis
  - Categorize requirements into essential and optional
  - Identify any potential conflicts or ambiguities
- Requirement specification
  - Prepare a brief SRS outline including system overview, actors and major features
- Requirement validation
  - Propose validation methods such as review or walkthrough
- Requirement management
  - Suggest how future changes (new feedback parameters, analytics features) can be managed

Submission Requirements –

Students must submit a single PDF or neatly written lab record containing –

- Case study understanding summary (5-6 lines)
- Feasibility study points (technical & operational)

- List of functional and non-functional requirements
- Categorized requirements (essential/ optional)
- Brief SRS outline
- Proposed validation method
- Requirement management strategy

Observations:

- Requirement engineering forms the base of SDLC
- Clear requirements reduce rework and development cost
- Stakeholder involvement is crucial during elicitation
- SRS acts as a contract between client and developer
- Requirement changes must be carefully controlled

Conclusion:

The experiment provided an understanding of requirement engineering tasks and their role in building successful software systems. Proper execution of requirement engineering activities ensures clarity, minimizes ambiguity and improves overall software quality

Expected Oral Questions:

1) What is requirement engineering?
2) Why is requirement engineering important?
3) List the tasks involved in requirement engineering
4) What is software requirement specification (SRS)?
5) Differentiate between functional and non-functional requirements
6) What techniques are used for requirement elicitation?
7) What is requirement validation?
8) Explain requirement management

FAQs in Interview:

1) What is requirement engineering?
2) Why is software requirement specification (SRS) important?
3) What happens if requirements are unclear?
4) Name common requirement elicitation techniques
5) What is requirement creep?